A Ternary Obstruction to Bounded-Odd-Step Inverse Collatz Searches

Collatz proof project

September 5, 2026

Abstract

For every fixed odd-step budget K, every positive-length shortcut Collatz segment ending at a target congruent to one modulo 3^K and containing at most K odd steps has a contracting final coefficient. The total segment length is unrestricted. Consequently no seed whose coefficients never contract can reach that target within the budget. We prove this in Lean using a sharp canonical-residue endpoint bound. Arbitrarily large targets outside multiples of three have this obstruction, but none is asserted to be a noncontracting seed. The result limits a proposed backward exclusion method; it does not prove Collatz. No priority claim is made.

1 The search question

Let $T(n) = n/2$ for even n and $(3n + 1)/2$ for odd n. Let $j_t(n)$ count odd states before time t. A seed satisfies NeverContracts(n) when $2^t \le 3^{j_t(n)}$ for every t. The preceding noncontracting-tail theorem proves that every unbounded positive orbit has a tail in this class, and that every positive seed in this class is unbounded.

A possible exclusion method is to take a least positive seed in the class and construct a smaller predecessor that is also in the class. Backward transport works through a prefix whose coefficients all stay at least one. The present theorem shows why a fixed bound on the number of odd steps in such constructions leaves a whole residue class unresolved, regardless of how many even steps are permitted.

2 A sharp endpoint bound

Theorem 1. *For every t and every $0 \le r < 2^t$,*

$$T^t(r) < 3^{j_t(r)}.$$

Proof. Induct on t using binary residue lifts. At depth t, write $e = T^t(r)$ and $A = 3^{j_t(r)}$, with $e < A$. Adding the next binary digit replaces r by $r + 2^t\varepsilon$, where $\varepsilon \in \{0, 1\}$. The exact affine transport identity gives the intermediate value $p = e + A\varepsilon < 2A$, with unchanged first-$t$ odd count. If p is even, its next value $p/2$ is below A. If p is odd, then $p \le 2A - 1$ and $(3p + 1)/2 \le 3A - 1 < 3A$; the odd count increases by one. This proves the next-depth bound. At depth zero the only residue is zero. □

3 The residue-one obstruction

Theorem 2. *If $t > 0$ and $2^t \le 3^{j_t(n)}$, then*

$$T^t(n) \not\equiv 1 \pmod{3^{j_t(n)}}.$$

Proof. Put $r = n \bmod 2^t$ and $A = 3^{j_t(n)}$. The odd-count transport theorem identifies $j_t(r)$ with $j_t(n)$, while the shadow congruence identifies the endpoints modulo A. If the displayed congruence held, then Theorem 1 and $A \ge 2^t \ge 2$ would force $T^t(r) = 1$.

The exact correction identity gives

$$2^t = Ar + d_t(r) \ge 2^t r,$$

so $r \le 1$. The zero seed cannot reach one, hence $r = 1$. But a positive return $T^t(1) = 1$ has strictly contracting coefficient by the existing cycle inequality: $A < 2^t$. This contradicts the hypothesis. $\square$

Theorem 3. *Suppose* $y \equiv 1 \pmod{3^K}$, $t > 0$, $j_t(n) \le K$, *and* $T^t(n) = y$. *Then*

$$3^{j_t(n)} < 2^t.$$

In particular, n *cannot satisfy* NeverContracts(n).

Proof. Since $3^{j_t(n)}$ divides 3^K, the target is also one modulo the smaller ternary scale. If the coefficient did not contract, Theorem 2 would give a contradiction. An infinitely noncontracting seed would in particular have a noncontracting coefficient at time t. $\square$

The number of even steps is unrestricted. The odd-step budget matters: the segment $3 \to 5 \to 8 \to 4$ has two odd steps and coefficient $9/8 > 1$, although its endpoint is one modulo three. It is outside the $K = 1$ budget. The positive-length hypothesis also matters; an identity segment need not contract.

4 Arbitrarily large unresolved targets

For any K and any height B, choose

$$y = 1 + 3^{K+1}(B + 1).$$

Then $y > B$, $y \equiv 1 \pmod 3$, and $y \equiv 1 \pmod{3^K}$. Theorem 3 obstructs every proposed noncontracting predecessor segment with at most K odd steps. Lean proves this existential statement for all K, B, so the obstruction cannot be removed by excluding just a fixed finite set of small targets.

This is a restriction on the backward method, not evidence of an escaping orbit. The constructed targets may all converge; no target is proved to satisfy NeverContracts. Other arguments could exclude the residue class directly. The result also does not exclude inverse methods whose odd-step budget grows with the target, or methods that establish noncontraction by a different argument.

5 Formal verification and next obligation

Source: `lean/Collatz/InverseBarrier.lean`. The main declarations are `canonical_endpoint_lt_three_pow`, `noncontracting_endpoint_not_one_mod`, `bounded_odd_inverse_obstruction`, and `arbitrarily_large_inverse_obstructions`. The module build and selected 144-declaration axiom audit pass using only standard foundations. No custom axiom or `sorry` is added. Build with `lake build Collatz.InverseBarrier` from `lean/`.

The proposed uniform bounded-odd-step predecessor coverage is therefore unavailable on this residue class. A full proof still needs an exclusion of all positive noncontracting seeds and an exclusion of nontrivial cycles. Neither is supplied here. The Collatz conjecture remains unresolved.